DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Tutorial Sheet 11: Solutions **4th and 5th May 2023**

1. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a continuous function and let D be a domain in $\mathbb{R}^2$ such that

$$\iint_D f(x,y)dxdy = \int_0^2 \int_{y^3}^{y+6} f(x,y)dxdy\,. \qquad (*)$$

(a) Let us first describe the domain D from the limits of the integrals in the right-hand side of $(*)$. Clearly the arc of curve $x = y^3$ with $y \in [0,2]$ and the segment line $x = y+6$ with $y \in [0,2]$ are involved in the description of the domain. They intersect at the point $(8,2)$. Therefore, D is the domain enclosed by curve $x = y^3$ and the line $x = y+6$ and the postive x-axis.

Sketch the domain D!

(b) Change the order of integration in $(*)$. **(Your Sketch is crucial here!)**

$$\boxed{\iint_D f(x,y)dxdy = \int_0^6 \left(\int_0^{x^{1/3}} f(x,y)dy\right) dx + \int_6^8 \left(\int_{x-6}^{x^{1/3}} f(x,y)dy\right) dx}\,.$$

2. **(Change of variables in double integrals)**

$$\boxed{\iint_D f(x,y)dxdy = \iint_R f(x(s,t),y(s,t))\big|J_T(s,t)\big|dsdt} \qquad (**)$$

where $T : \mathbb{R}^2 \to \mathbb{R}^2$ is invertible with T, T^{-1} having continuous partial derivatives and Jacobian J_T different from 0 also $(x,y) = T(s,t)$.

(a) Evaluate $\displaystyle\iint_D xydxdy$ where D is the parallelogram bounded by the lines $x - 3y = 1$, $x - 3y = 3$, $2x + y = -2$ and $2x + y = 0$.

The description of the domain clearly suggests the use of a change of variables for the evaluation of the double integral. Using the notation in the notes, we set

$$\begin{cases} s = x - 3y \\ t = 2x + y \end{cases} \quad \Rightarrow \quad T : \begin{cases} x = \dfrac{1}{7}(s+3t) \\[2ex] t = \dfrac{1}{7}(-2s+t) \end{cases} \qquad \text{with} \quad |J_T| = \frac{1}{7}\,.$$

Hence, $D = T(R)$ where

$$R := \{(s,t) \in \mathbb{R}^2 \colon\ 1 \le s \le 3 \text{ and } -2 \le t \le 0\} = [1,3] \times [-2,0]\,.$$

So, we obtain that

$$\begin{aligned}
\iint_D xy\,dxdy &= \iint_R \frac{1}{49}(s+3t)(-2s+t)\frac{1}{7}dsdt = \frac{1}{343}\int_1^3 \left(\int_{-2}^0 (s+3t)(-2s+t)dt\right) ds \\
&= \frac{1}{343}\int_1^3 \left(\int_{-2}^0 (-2s^2 - 5st + 3t^2)dt\right) ds = \frac{1}{343}\int_1^3 \left[-2s^2t - \frac{5}{2}st^2 + t^3\right]_{-2}^0 ds \\
&= \frac{1}{343}\int_1^3 (-4s^2 + 10s + 8)ds = \frac{1}{343}\left[-\frac{4}{3}s^3 + 5s^2 + 8s\right]_1^3 = \frac{64}{343 \times 3}
\end{aligned}$$

(b) Let $D := \{(x,y) \in \mathbb{R}^2 \colon\ 0 \le x \le 2, \quad 0 \le x - y \le 2\}$.

Evaluate the double integral $\displaystyle\iint_D \frac{(x-y)^2}{1+(x-y)^2}\,dxdy\,.$

From the description of the domain D, we can use the change of variables

$$\begin{cases} s = x, \\ t = x - y \end{cases} \quad \Rightarrow \quad T : \begin{cases} x = s, \\ y = s - t, \\ (s,t) \in [0,2] \times [0,2] \end{cases}$$

for which

$$J_T(s,t) = \begin{vmatrix} x_s(s,t) & x_t(s,t) \\ y_s(s,t) & y_t(s,t) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 1 & -1 \end{vmatrix} = -1\,.$$

Thus

$$\begin{aligned}
\iint_D \frac{(x-y)^2}{1+(x-y)^2}\,dxdy &= \int_0^2 \int_0^2 \frac{t^2}{1+t^2}|-1|dsdt = \int_0^2 \left(\int_0^2 \left(1 - \frac{1}{1+t^2}\right)dt\right) ds \\
&= \int_0^2 \Big[t - \arctan t\Big]_0^2 ds = 2(2 - \arctan 2)\,.
\end{aligned}$$

Notice that, one could also evaluate this double inegral by writing

$$D := \{(x,y) \in \mathbb{R}^2 \colon\ 0 \le x \le 2, \quad x - 2 \le y \le x\}$$

and get

$$\begin{aligned}
&\iint_D \frac{(x-y)^2}{1+(x-y)^2}\,dxdy = \int_0^2 \int_{x-2}^x \left(1 - \frac{1}{1+(x-y)^2}\right)dydx \\
&= \int_0^2 \Big[y + \arctan(x-y)\Big]_{x-2}^x dx = \int_0^2 \big(x - (x - 2 + \arctan 2)\big)dx \\
&= \int_0^2 (2 - \arctan 2)dx = 2(2 - \arctan 2)\,.
\end{aligned}$$

3. **Evaluation of double integral using polar coordinates:** We recall that if $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $(x, y) = T(r, \theta)$ with $x = r\cos\theta$ and $y = r\sin\theta$ and $D = T(R)$ then

$$\iint_D f(x,y)dxdy = \iint_R f(r\cos\theta, r\sin\theta) r dr d\theta\,.$$

Evaluate each of the following integrals:

(i) $\displaystyle\iint_D \frac{1}{1+x^2+y^2}dxdy$ where $D := \{(x,y) \in \mathbb{R}^2\colon\ x^2+y^2 \le 4,\ y \le 0\}$;
Using the change of variables in polar coordinates $x = r\cos\theta$ and $y = r\sin\theta$ we find that

$$\iint_D \frac{1}{1+x^2+y^2}dxdy = \int_\pi^{2\pi}\int_0^2 \frac{r}{1+r^2}drd\theta = \pi\Big[\frac{1}{2}\ln(1+r^2)\Big]_0^2 = \frac{1}{2}\pi\ln 5\,.$$

(ii) $\displaystyle\iint_D \frac{x}{x^2+y^2}dxdy$ where $D := \{(x,y) \in \mathbb{R}^2\colon\ 0 \le y \le x,\ x^2+y^2 \ge 1,\ x \le 1\}$.
Using the change of variables in polar coordinates $x = r\cos\theta$ and $y = r\sin\theta$ we find that

$$0 \le y \le x \quad\Rightarrow\quad 0 \le \tan\theta \le 1 \quad\Rightarrow\quad 0 \le \theta \le \frac{\pi}{4}\,.$$

On the other hand,

$$x^2+y^2 \ge 1 \quad \text{and} \quad x \le 1 \quad\Rightarrow\quad 1 \le r \le \frac{1}{\cos\theta}\,.$$

Therefore,

$$\iint_D \frac{x}{x^2+y^2}dxdy = \int_0^{\frac{\pi}{4}}\int_1^{\frac{1}{\cos\theta}} \frac{r\cos\theta}{r^2} r dr d\theta = \int_0^{\frac{\pi}{4}}(1-\cos\theta)d\theta = \big[\theta - \sin\theta\big]_0^{\frac{\pi}{4}} = \frac{\pi}{4} - \frac{\sqrt{2}}{2}$$

4.

$$R = \{(x,y,z)\colon\ -1 \le y \le 1,\ 0 \le x \le 1-y^2,\ -\sqrt{x} \le z \le \sqrt{x}\}$$

$$\begin{aligned}\iiint_R 2y^2\sqrt{x}dxdydz &= \int_{-1}^{1}\left[\int_0^{1-y^2}\left[\int_{-\sqrt{x}}^{\sqrt{x}} dz\right]\sqrt{x}dx\right]2y^2dy = \int_{-1}^{1}\left[\int_0^{1-y^2} 2xdx\right]2y^2dy \\ &= \int_{-1}^{1} 2(1-y^2)^2y^2dy = 4\int_0^1 y^2(1-y^2)^2dy = \frac{32}{105}\end{aligned}$$

5. Let $a > 0$, $b > 0$ and $c > 0$. If T is the tetraedron bounded by the three coordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ then show by means of a triple integral that the volume of T is equal to $\frac{abc}{6}$.
T can be described as the solid lying below the graph of the function $z = c(1 - \frac{x}{a} + \frac{y}{b})$

defined on the domain D on the xy plane, enclosed by the x-axis, the y-axis and the line $\frac{x}{a}+\frac{y}{b}=1$. Hence, the volume of T is

$$\begin{aligned} V(T) &= \iint_D c\Big(1-\frac{x}{a}+\frac{y}{b}\Big)dxdy = \int_0^a \int_0^{b(1-\frac{x}{a})} c\Big(1-\frac{x}{a}-\frac{y}{b}\Big)dydx \\ &= c\int_0^a \Big[\big(1-\frac{x}{a}\big)y - \frac{y^2}{2b}\Big]_0^{b(1-\frac{x}{a})} dx = \frac{bc}{2}\int_0^a \big(1-\frac{x}{a}\big)^2 dx \\ &= \frac{bc}{2}\Big[-\frac{a}{3}\big(1-\frac{x}{a}\big)^3\Big]_0^a = \frac{abc}{6}. \end{aligned}$$

6. Convert the integral $\int_{-2}^{2}\int_0^{\sqrt{4-y^2}}\int_0^x (x^2+y^2)dzdxdy$ into an equivalent integral in cylindrical coordinates and evaluate the result.

   This is the triple integral of the function $f(x,y) := x^2+y^2$ over the region enclosed by the half cylinder $x^2+y^2=4$ with $x \geq 0$ and the planes $x=0$, $z=0$ and $z=x$. So, the tiple integral is converted in cylindrical coordinates as

$$\begin{aligned} \int_{-2}^{2}\int_0^{\sqrt{4-y^2}}\int_0^x (x^2+y^2)dzdxdy &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^2\int_0^{r\cos\theta} r^3 dzdrd\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^2 r^4\cos\theta drd\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Big[\frac{1}{5}r^5\Big]_0^2 \cos\theta d\theta = \frac{32}{5}\Big[\sin\theta\Big]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{64}{5}. \end{aligned}$$

7. Let E be the region above the cone $z=\sqrt{x^2+y^2}$ and inside the sphere $x^2+y^2+z^2=4$. Evaluate the triple integral $\iint_E (x^2+y^2)zdxdydz$ through:

   (a) cylindrical coordinates: $x = r\cos\theta$, $y = r\sin\theta$ and $z=z$.

   The sphere is described in cylindrical coordinates as $r^2+z^2=4$ while the cone is described as $z=r$.
   We first find the intersection of the cone and the sphere to get

$$\begin{cases} z=\sqrt{x^2+y^2} \\ \underbrace{x^2+y^2}_{=z^2}+z^2=4 \end{cases} \Leftrightarrow \begin{cases} z=\sqrt{x^2+y^2} \\ 2z^2=4 \end{cases} \Leftrightarrow \begin{cases} x^2+y^2=2 \\ z=\sqrt{2} \end{cases}$$

   Hence, the region E is described in cylindrical coordinates r, θ, z as:

$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq \sqrt{2}, \quad r \leq z \leq \sqrt{4-r^2}.$$

   Now, recalling that the Jacobian when using of the cylindrical coordinates is equal to

r, we find that

$$\iiint_E (x^2+y^2)z\,dxdydz = \int_0^{2\pi}\int_0^{\sqrt{2}}\int_r^{\sqrt{4-r^2}} r^2 zr\,dzdrd\theta = \int_0^{2\pi}\int_0^{\sqrt{2}} r^3\left[\frac{1}{2}z^2\right]_r^{\sqrt{4-r^2}} drd\theta$$
$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\sqrt{2}} r^3(4-r^2-r^2)drd\theta = \frac{1}{2}\int_0^{2\pi}\int_0^{\sqrt{2}}(4r^3-2r^5)drd\theta = \frac{1}{2}\int_0^{2\pi}\left[r^4-\frac{1}{3}r^6\right]_0^{\sqrt{2}} d\theta$$
$$= \frac{1}{2}\int_0^{2\pi}(4-\frac{8}{3})d\theta = \frac{4\pi}{3}\,.$$

(b) spherical coordiantes.

Notice that

$$x=\rho\sin\phi\cos\theta,\quad y=\rho\sin\phi\sin\theta,\quad z=\rho\cos\phi \quad \text{with Jacobian} \quad J_T=\rho^2\sin\phi\,.$$

The sphere is described in spherical coordinates as $\rho=\sqrt{x^2+y^2+z^2}=2$ while the cone is described as

$$z=\sqrt{x^2+y^2} \Leftrightarrow \rho\cos\phi=\rho\sin\phi \Leftrightarrow \cos\phi=\sin\phi \Leftrightarrow \tan\phi=1 \Leftrightarrow \phi=\frac{\pi}{4}\,.$$

Thus, the region E is described in spherical coordinates as

$$0\le\theta\le 2\pi,\quad 0\le\rho\le 2,\quad 0\le\phi\le\frac{\pi}{4}\,.$$

So,

$$\iiint_E (x^2+y^2)z\,dxdydz = \int_0^{2\pi}\int_0^{2}\int_0^{\frac{\pi}{4}}(\rho^2\sin^2\phi)(\rho\cos\phi)(\rho^2\sin\phi)d\phi d\rho d\theta$$
$$= \int_0^{2\pi}\int_0^{2}\rho^5\left(\int_0^{\frac{\pi}{4}}\sin^3\phi\cos\phi d\phi\right)d\rho d\theta = \int_0^{2\pi}\int_0^{2}\rho^5\left[\frac{1}{4}\sin^4\phi\right]_0^{\frac{\pi}{4}} d\rho d\theta$$
$$= \int_0^{2\pi}\int_0^{2}\frac{1}{16}\rho^5 d\rho d\theta = \frac{1}{16}\int_0^{2\pi}\left[\frac{1}{6}\rho^6\right]_0^2 d\theta = \frac{2\pi\times 64}{16\times 6} = \frac{4\pi}{3}\,.$$